Cost-Aware Reinforcement Learning for Agentic RAG

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Abstract

Tools help on some questions and hurt on others, so the average can stay flat. We price retrieve, rewrite, rerank, verify, and stop inside one retrieval episode, and we train a 357-parameter controller with REINFORCE. The reward subtracts token dollars, retrieval dollars, a fixed latency table, and a charge per action from answer quality. On 150 HotpotQA items and 150 Natural Questions items, with BM25 over 83,120 passages, one-shot retrieval scores 60/150 and 40/150. Max tools scores 62/150 and 43/150. That is 105 correct against 100, and the reward falls from 0.581 to 0.503. Three seeds at dollar weight 20 each select rewrite, then rewrite, then retrieve, then stop. The recipe scores 52/150 and 48/150, again 100 correct. It repairs 5 HotpotQA answers and breaks 13, and repairs 17 Natural Questions answers and breaks 9. The union of those two is 122. An oracle over one-shot retrieval, max tools, and this recipe scores 126/300 at 1.81×10^{-4} dollars. The branch on retrieval strength appears in 0 of 3 seeds.

1 Introduction

On the Natural Questions item "what is the name of wonder womans mother," one-shot retrieval returns passages titled Seoul Korea Temple and Flash in other media, and the generator answers Stevie Wonder. Two rewrites, together 2.4×10^{-5} dollars, change the query to "mother of wonder woman." The next search returns Wonder Girl and Duke of Deception, and the answer is Queen Hippolyta. The episode costs 2.06×10^{-4} dollars, against 1.83×10^{-4} for one-shot retrieval. On this question the rewrite pays. On others it does not.

That is the story. Tools help per question, not on average. The reward decides which tools survive. A small controller can lock onto one recipe and still miss the per-question choice.

Figure 1 is the whole pipeline. The generator and the retriever are frozen. Every number below is on one index of 83,120 passages.

- The reward decides which tools survive. Max tools scores 62/150 on HotpotQA and 43/150 on Natural Questions, against 60/150 and 40/150 for one-shot retrieval. Five more answers, and the reward falls from 0.581 to 0.503, because the action charge rises by 0.10 and the weighted quality terms rise by about 0.025.
- The help is per question. The learned recipe and one-shot retrieval each score 100/300. The recipe is 52/150 and 48/150. One-shot retrieval is 60/150 and 40/150. The recipe repairs 5 HotpotQA answers and breaks 13, and repairs 17 Natural Questions answers and breaks 9. The union is 122/300. An oracle over one-shot retrieval, max tools, and the learned recipe, keeping the cheapest correct trajectory, scores 126/300 at 1.81×10^{-4} dollars. The six-step recipe adds no question one-shot retrieval missed.
- The controller finds a stable recipe and does not learn that choice. Seeds 42, 43, and 44 at λ=20 each emit rewrite, rewrite, retrieve, stop on all 300 questions. The second action still sees a retrieval score of 0. The branch on retrieval strength appears in 0 of 3 seeds.

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2 Related Work

Retrieval-augmented generation conditions a generator on retrieved text (Lewis et al. 2020). Agentic controllers interleave that retrieval with other tools. ReAct alternates reasoning traces and actions (Yao et al. 2023). The five systems in Table 1 are the ones a reader will set this paper against. Each decides something about retrieval. None prices rewrite, rerank, and verify in dollars inside a closed loop.

Search-R1 trains the language model with an outcome reward to issue searches and reason over the results (Jin et al. 2025). GRASP is the closest typed-action neighbor: a GRPO policy over semantic search, keyword search, and paragraph reading, rewarded for answer F1, grounded reading, complementary search, and turn efficiency (Gandhi et al. 2026). The efficiency term is paid only when answer F1 exceeds 0.5, and it counts turns, not dollars. GRASP does not price rewrite, rerank, or verification. We keep the generator fixed and ask a prior question: once those tools have a dollar cost, does a learned controller spend them only when they pay?

FrugalGPT and RouteLLM price a different choice. FrugalGPT cascades queries across LLM APIs whose fees differ by orders of magnitude, so a cheap model answers when it can and an expensive model answers when it must (Chen et al. 2023). RouteLLM learns a router from preference data and sends each query to a strong model or a weak one (Ong et al. 2024). Both decide which model to call. We

Method	What it decides	What it prices	What we add
Adaptive-RAG (Jeong et al. 2024)	No retrieval, one retrieval, or multi-step retrieval, from the question alone	Step count and relative time	A second look after the passages arrive, with dollars and a per-action charge
Self-RAG (Asai et al. 2024)	Whether to retrieve, and whether a segment is supported, as tokens in the generator	Hand-set critique weights	A frozen generator. Critique is one typed action with a measured cost
CRAG (Yan et al. 2024)	Keep, discard, or mix the retrieved set. Discard triggers a web search	An evaluator threshold	Rewrite, rerank, and verify on a closed corpus
Search-R1 (Jin et al. 2025)	When the language model itself emits a search during reasoning	Answer correctness	A small controller, so a change in exact match is a change in control
GRASP (Gandhi et al. 2026)	Semantic search, keyword search, or a paragraph read	Answer quality, grounded reading, using both searches, and a turn count	Rewrite, rerank, and verify, priced in dollars

Table 1: Published controllers this paper is positioned against. None is reimplemented on this stack. Adaptive-RAG is the row that could be: it is a classifier in front of no retrieval, one-shot retrieval, and a capped multi-step loop, all of which this environment already runs. That classifier is not trained here, so Adaptive-RAG is not a row in the results.

use one generator, Qwen2.5-3B-Instruct, and price the tool calls inside the episode.

The retrieval line behind that generator is older than the agent. REALM trains a retriever inside language-model pre-training (Guu et al. 2020). Fusion-in-Decoder reads each retrieved passage on its own and fuses them in the decoder (Izacard and Grave 2020). RETRO conditions each chunk of the generation on neighbors from a trillion-token store (Borgeaud et al. 2021). Atlas asks how far a retrieval-augmented model can go in the few-shot regime (Izacard et al. 2022).

Later controllers decide when to search again. IRCoT interleaves retrieval with chain-of-thought steps (Trivedi et al. 2022). FLARE retrieves when the next sentence looks uncertain (Jiang et al. 2023). DRAGIN triggers retrieval from the model's own information need (Su et al. 2024). Self-Ask splits a multi-hop question into follow-ups (Press et al. 2022). Toolformer inserts API calls, including a search engine, into the generation (Schick et al. 2023). On the query side, Ma et al. rewrite, then retrieve, then read (Ma et al. 2023). HyDE embeds a hypothetical answering passage instead of the question (Gao et al. 2022). monoBERT reranks BM25 hits with a cross-encoder (Nogueira and Cho 2019). RankGPT asks a language model to permute the candidates (Sun et al. 2023).

The training line splits the same way. PPO is the usual clipped policy gradient (Schulman et al. 2017). GRPO, introduced for math reasoning, replaces the learned critic with a group of rollouts of the same prompt (Shao et al. 2024). R1-Searcher and ReSearch use that style of update to teach a language model to search (Song et al. 2025; Chen et al. 2025). RARR looks up evidence and revises a statement that the evidence does not support (Gao et al. 2022). TRUE compares the metrics people use for that kind of check (Honovich et al. 2022). We do not reimplement these systems. They are the decisions our five actions stand in for.

3 Method

Figure 1 is the protocol. An episode is one question. The controller picks one action at each step. The episode ends at stop, or when the step or budget cap forces a stop.

Actions, Retriever, and Generator

The action set is retrieve, rewrite, rerank, verify, and stop. Retrieve runs BM25 (Robertson and Zaragoza 2009) and appends the top five passages. Rewrite asks the generator for a new query. Rerank reorders the current pool by lexical overlap with the query. Verify runs a lexical natural-language-inference check of the current draft and returns support, neutral, or contradiction. Stop emits a short answer, or abstains. Yes/no questions must be answered yes or no. Caps are eight steps, three retrieves, two rewrites, two reranks, and two verifies. Illegal actions are masked: with no evidence the policy cannot rerank, verify, or stop.

The generator is Qwen2.5-3B-Instruct at temperature 0 (Qwen Team 2024). Decoding is greedy. It reads the passages. The controller does not. The verifier is lexical, not a cross-encoder and not the generator judging itself. Training ran on one NVIDIA T4. The policy itself is a CPU network.

The price card meters local tokens at \$0.15 and \$0.60 per million input and output tokens, and BM25 at 5×10^{-5} dollars per call. Latency in the reward and in the budget check is a fixed table, not the wall clock: 212.4 ms per retrieve, 999.2 ms per rewrite, 143.2 ms per rerank, 0 ms per verify, and 853.9 ms plus 52.0 ms per completion token for a generation. Those values are the median measured cost of the same calls, so the term keeps its size. A wall-clock charge would make two episodes with the same actions differ by a few milliseconds, and that difference enters the return. The table removes it. The logged latency is still the measured time, which is why the three exams below can match on every prediction and still disagree on milliseconds. The latency weight is 0.5 and the unit is 10^{-3} dollars per second. Absolute dollars are an

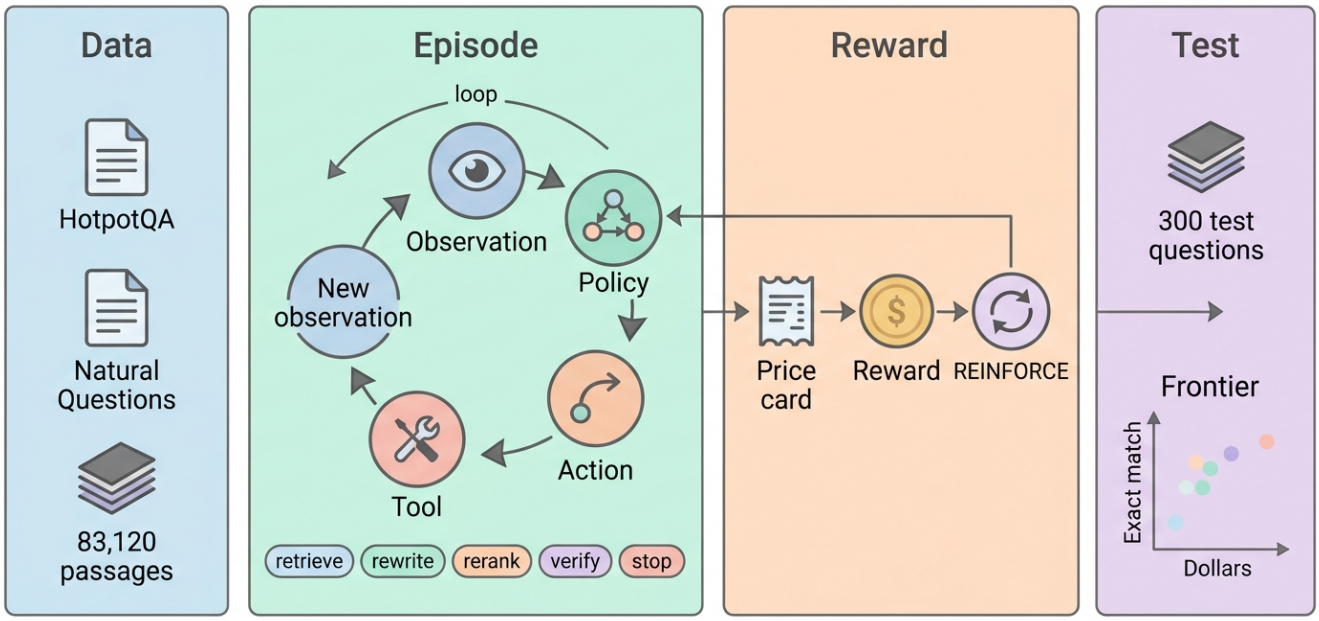


Figure 1: The episode. HotpotQA and Natural Questions share an index of 83,120 passages. The controller loops from an observation through one of five actions and back. The price card feeds the reward, REINFORCE carries that reward to the policy, and the 300 test questions feed the frontier.

instrumentation scale for a local model. Ratios are what the policy can learn.

Observation, Training, and Selection

The observation is 16 numbers. Four of them are retrieval scores: the mean of the top five, the top score, the gap between the top two, and the minimum of the top five. Each is a z-score against the training slice, clipped to $[-3, 3]$. The rest are the evidence count, remaining steps, remaining budget, verifier support and contradiction, the four tool counts, and a verify one-hot for support, contradiction, and neutral. There is no dataset indicator and no passage text. Before any retrieval the score channels are 0, the budget is full, and the vector does not contain the question. Every question therefore presents the same observation at step zero, and the first action cannot depend on which question is being asked. Adaptivity begins at the second step, after a retrieval has been observed.

The policy is a multilayer perceptron with one hidden layer of width 16, which is 357 parameters. Width above 32 is rejected. We train with REINFORCE (Williams 1992) for 20 epochs. The return is the terminal reward below. The advantage on a training question is that question's sampled reward minus the reward of one-shot retrieval on the same question. The entropy bonus coefficient is 0.01, the optimizer is Adam, and the learning rate is 0.003. Training uses 500 questions, 300 from HotpotQA and 200 from Natural Questions. After each epoch we roll out the mode on a held-out 180, 110 HotpotQA and 70 Natural Questions. The checkpoint we report is the first epoch that attains the highest greedy reward on that 180. It is chosen before the test set is opened. The test action is the mode of the masked softmax.

The exam below is three trainings at λ=20 and ρ=0, seeds 42, 43, and 44. We report the three exams, not a mean and a standard deviation. The generator seed is temperature 0. The policy seed sets the initial weights and the per-question action sample during training.

Reward

The terminal reward is

$$\begin{aligned} R = {} & \alpha Q_{\text{ans}} + \beta Q_{\text{ground}} + \gamma Q_{\text{cal}} \\ & - \lambda(C_{\text{tok}} + C_{\text{ret}}) - C_{\text{lat}} \\ & - P_{\text{hall}} - P_{\text{act}} - P_{\text{bud}} + B. \end{aligned} \tag{1}$$

Q_{ans} is the average of exact match and token F1, the two scores defined for SQuAD (Rajpurkar et al. 2016). Q_{ground} mixes lexical overlap between the answer and the passages with, on HotpotQA, recall of the gold supporting titles supplied with that dataset (Yang et al. 2018). C_{tok} and C_{ret} are price-card dollars. C_{lat} is the table latency times the latency weight. P_{hall} penalizes unsupported and contradicted claims. P_{bud} is a unit penalty if a hard budget breaks. The unused-budget bonus B is our term. GRASP pays its turn-efficiency bonus only when answer F1 exceeds 0.5, so a short wrong answer does not collect it (Gandhi et al. 2026). We use that gate, $Q_{\text{ans}} \geq 0.5$, and we pay the bonus on remaining budget rather than on unused turns. The action charge is $P_{\text{act}} = \rho \max(0, n-1)$ for n actions in the episode. Calibration, the hallucination penalty, the action charge, and the dollar weights are ours.

Default weights are α=1, β=0.4, γ=0.15, λ=2, latency weight 0.5, hallucination weight 0.5, and ρ=0.02. The three-seed exam sets ρ=0 and λ=20. A correct answer scores

$+0.3$ on the calibration term. Abstaining scores $+0.6$ when the passage list is empty, the mean retrieval score is below 3, or verify returns contradiction, and -0.2 otherwise. On this corpus the mean-score arm does not fire: raw BM25 means stay above 20, and means after the lexical reranker stay above 7. Empty evidence and a contradiction label still earn $+0.6$. A wrong answer scores -0.4 with evidence and -0.2 without it.

Data

The test set is locked at 300 questions: 150 HotpotQA distractor items (Yang et al. 2018) and 150 Natural Questions items (Kwiatkowski et al. 2019), disjoint from training and validation. The index holds HotpotQA contexts, unused HotpotQA distractors, and the DPR 100-word Wikipedia passages released with Natural Questions (Karpukhin et al. 2020; Gao et al. 2022), including the passages needed by the 500 training and 180 validation questions, 83,120 passages in total. Gold text is a Wikipedia passage. BM25 recall at five, measured on the 80,000-passage snapshot of this evaluation set, is 0.927 on HotpotQA and 0.587 on Natural Questions, so a second retrieval can matter. Three frozen controllers define the corners of the cost plot: one-shot retrieval, a rule that reranks and verifies when the mean score is at least 3, and a max-tool policy that spends three retrieves, one rewrite, one rerank, and one verify. A fourth frozen string, retrieve then rewrite then rewrite then verify then verify then stop, is the six-step recipe used in the ceilings.

4 Results

Extra Tools Buy Five Answers and Lose Reward

Table 2 is the frozen controllers on the 83,120-passage index. One-shot retrieval scores 100/300. Using every tool scores 105/300. From one-shot retrieval to max tools, Q_{ans} rises by 0.019, Q_{ground} by 0.012, and Q_{cal} by 0.010, about 0.025 once the weights in Equation 1 are applied. The action charge rises from 0.02 to 0.12, because the episode grows from 2 actions to 7 and ρ=0.02. The reward falls from 0.581 to 0.503. One extra retrieval costs roughly 3×10^{-4} in the λ=2 dollar term and 0.02 in ρ, so the flat charge is about sixty times the measured dollar price of that call. The rule policy pays for a rerank and a verify on every question, lands between them on cost, and scores 97/300. Table 3 is the same four controllers under the default weights, λ=2 and ρ=0.02. Max tools has the highest answer quality and the lowest reward. The six-step recipe matches one-shot retrieval on Q_{ans} and Q_{cal} and spends more, so its reward is the lowest of the four. The learned exam below uses λ=20 and ρ=0, so its return is not a row of this table.

The Ceiling on This Index Is 126

Figure 2 plots exact match against dollars. An oracle keeps the cheapest correct trajectory among one-shot retrieval, max tools, and the learned recipe, and the cheapest trajectory when none of them is correct. The six-step recipe emits the same 300 answers as one-shot retrieval, so it adds no question. One-shot retrieval and the learned recipe cover 122 questions between them. Max tools adds 4 questions that both of those miss. The oracle scores 126/300 at 1.81×10^{-4} dollars. Single policies span 97 to 105 correct, and the learned recipe scores 100. The room above one-shot retrieval, among trajectories we logged, is 26 answers. Assigning one-shot retrieval to every HotpotQA question and the six-step recipe to every Natural Questions question scores 100/300 at 3.29×10^{-4} dollars, because that recipe is more expensive and does not change an answer.

Three Seeds Select One Recipe

Table 4 is the λ=20 exam. Seeds 42, 43, and 44 select epochs 17, 14, and 11, the first epoch that attains the best greedy reward on the 180 validation questions. The three exams match on every prediction, every action string, exact match, and reward. The action string is rewrite, rewrite, retrieve, stop on all 300 questions. Exact match is 100/300, HotpotQA 52/150 and Natural Questions 48/150, at 2.02×10^{-4} dollars. The return under this preset, λ=20 and ρ=0, is 0.598 on each seed. That number is not comparable to the default-weight rewards in Table 3. Measured latency, which is logged and not priced, is 3061 ms, 3000 ms, and 2956 ms.

Against one-shot retrieval the recipe agrees on 179 of 300 predictions. Of the 121 disagreements, 22 become correct, 22 become incorrect, 1 stays correct under a different string, and 76 stay wrong. The repairs and the breaks are HotpotQA $+5$ and -13, Natural Questions $+17$ and -9. The union is 122/300. Both policies are right on 78 questions, so $78+22+22=122$. Figure 3 splits those 300 questions by dataset.

That union is the pairwise coverage of one-shot retrieval and the learned recipe. The ceiling in Figure 2 keeps the cheapest correct trajectory among one-shot retrieval, max tools, and this learned recipe. The six-step recipe adds no new correct question. Max tools adds 4 questions that one-shot retrieval and the learned recipe both miss, so the ceiling is 126/300 at 1.81×10^{-4} dollars. Any one policy still scores about 100. The questions move.

The recipe never branches on the retrieval score. We count a seed only when the second action is sometimes rewrite and sometimes retrieve, at least 15 times each, and one threshold on the mean score matches at least 250 of those questions. Here the first action is already rewrite, so the second action still sees a mean score of 0 and no passages. That action is rewrite on all 300 questions, on all three seeds. Retrieve is never the second action. The count is 0 of 3.

Verify is not what moves the 44 answers. The learned recipe never calls it. The six-step recipe calls it twice on every question and still emits the same 300 answers as one-shot retrieval.

Diagnostics

An earlier observation passed the mean BM25 score through $\tanh(\text{score}/5)$. Top-1 scores on this corpus run from about 26 to 122, so that feature was 1.0 on all 300 questions and a policy could not branch on it. Standardizing the score makes a branch expressible. The three seeds do not use it. The first action is chosen before any score exists, and the selected string spends both rewrites before the first retrieve. The earlier saturated feature is the reason the branch had to be defined, not a second result.

	One-shot	Rule	Max tools	Six-step recipe
HotpotQA	60/150	56/150	**62/150**	60/150
Natural Questions	40/150	41/150	**43/150**	40/150
Correct	100/300	97/300	**105/300**	100/300
$ × 10^{-4}	**1.72**	5.04	7.47	4.62
Reward	**0.581**	0.534	0.503	0.498

Table 2: Frozen controllers on the 83,120-passage index. Dollars are shown times 10^{-4}. Bold is the best value in the column: most correct answers, highest reward, or lowest dollars. The six-step recipe is retrieve, rewrite, rewrite, verify, verify, stop. It emits the same 300 answers as one-shot retrieval.

	Q_{ans}	Q_{ground}	Q_{cal}	$ × 10^{-4}	Reward
One-shot	0.366	0.646	−0.137	**1.72**	**0.581**
Rule	0.360	0.652	−0.146	5.04	0.534
Max tools	**0.385**	**0.658**	**−0.127**	7.47	0.503
Six-step recipe	0.366	0.646	−0.137	4.62	0.498

Table 3: Reward terms for the frozen controllers, under the default weights λ=2 and ρ=0.02. Dollars are shown times 10^{-4}. Bold is the best value in the column. Q_{ans} is the average of exact match and token F1. Max tools leads on quality and trails on reward. The learned exam uses a different weight vector and is not in this table.

5 Limitations

There are three limits on what these results support.

First, the controller never reads passage text. This is a study of the reward and of credit assignment in a small policy, not a fine-tune of the generator. The first action is the same on every question, because the observation before retrieval does not contain the question. The selected recipe rewrites twice before it retrieves, so the second action is still blind to the retrieval score. Adaptivity that depends on a score can start only after a retrieve.

Second, the three seeds are λ=20 with the action charge off. They are three values of one setting. They are not a sweep of λ, and they are not a standard deviation. The frozen rows in Table 2 are one pass each, at temperature 0.

Third, the verifier and the reranker are lexical, and the price card is a scale for a local model. A neural verifier, a dense retriever, or a generator-sized policy could move the 26-answer gap and the dollar weight at which extra tools stop paying. That comparison was not run. A question-only classifier in the style of Adaptive-RAG, trained on the 500 training questions to choose one-shot retrieval or the rewrite-first recipe, is also not in the tables. Scoring both recipes on those 500 questions needs the generator, and that pass was not run. Adaptive-RAG, Self-RAG, CRAG, Search-R1, and GRASP remain Table 1.

6 Conclusion

On one index, extra tools buy five exact-match answers and lose reward. Max tools scores 105/300 against 100/300 for one-shot retrieval, and the reward falls from 0.581 to 0.503 because the action charge rises by more than the quality terms.

The per-question ceiling among the trajectories we logged is 126/300, at 1.81×10^{-4} dollars, near the one-shot price, because the oracle pays for a more expensive trajectory only on the 26 answers one-shot retrieval missed. A frozen six-step recipe does not add a question. Three seeds of the learned policy select one four-step recipe, score 100/300, and disagree with one-shot retrieval on 44 questions. The branch on retrieval strength appears in 0 of 3 seeds, because that recipe rewrites before it ever sees a score.

A next controller has to read the question, or the passages, before it spends a rewrite. Three changes would test that. A neural verifier could replace the lexical check that currently changes no answer. A policy that reads the passage text could choose the second action from the evidence, not from a score of 0. GRPO, which compares several rollouts of the same question, is a closer trainer to GRASP than the one-sample REINFORCE update used here (Shao et al. 2024; Gandhi et al. 2026). None of the three was run.

AI Use

A language model was used to revise sentences and to check bibliography entries against paper abstracts. An image model

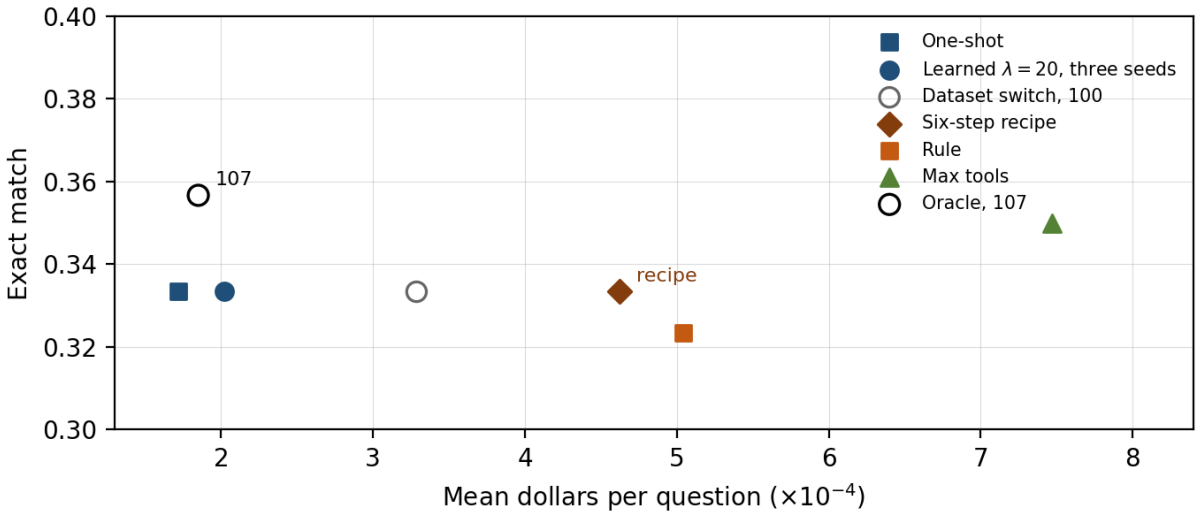


Figure 2: Exact match against mean dollars per question on the 83,120-passage index. Squares are one-shot retrieval and the rule. The triangle is max tools. The diamond is the six-step recipe. The filled circle is the learned λ=20 exam; seeds 42, 43, and 44 sit on the same point. The gray circle is the dataset switch, which scores 100. The open circle is the oracle over one-shot retrieval, max tools, and the learned recipe, which scores 126.

Seed	Epoch	HotpotQA	Natural Questions	Correct	Steps	$ × 10⁻⁴
42	17	52/150	48/150	100/300	4	2.02
43	14	52/150	48/150	100/300	4	2.02
44	11	52/150	48/150	100/300	4	2.02

Table 4: The three λ=20 exams. Each seed emits one action string, rewrite then rewrite then retrieve then stop. The rows tie on every reported value except the selected epoch, so no cell is marked as best. One-shot retrieval on the same index scores 100/300 at 1.72×10^{-4} dollars. We report the three rows, not a mean.

drafted Figure 1 and the four tables. The experimental plots were drawn from the logged numbers, and every table cell was checked against those numbers. No model output is reported as a measurement.

Ethical Statement

Pricing a tool can refuse a search that would have changed the answer, and the user is not told. Fixed recipes score 97 to 105 out of 300, and an oracle over one-shot retrieval, max tools, and the learned recipe reaches 126 near the one-shot price. The learned recipe and one-shot retrieval each score 100 and disagree on 44 questions. The abstention bonus does not fire on this corpus, and the selected checkpoints never call the verifier. The dollars are a scale for a local 3B model, not a bill a person pays. The questions and passages are public benchmarks with no human subjects, and the checkpoints are not a serving policy.

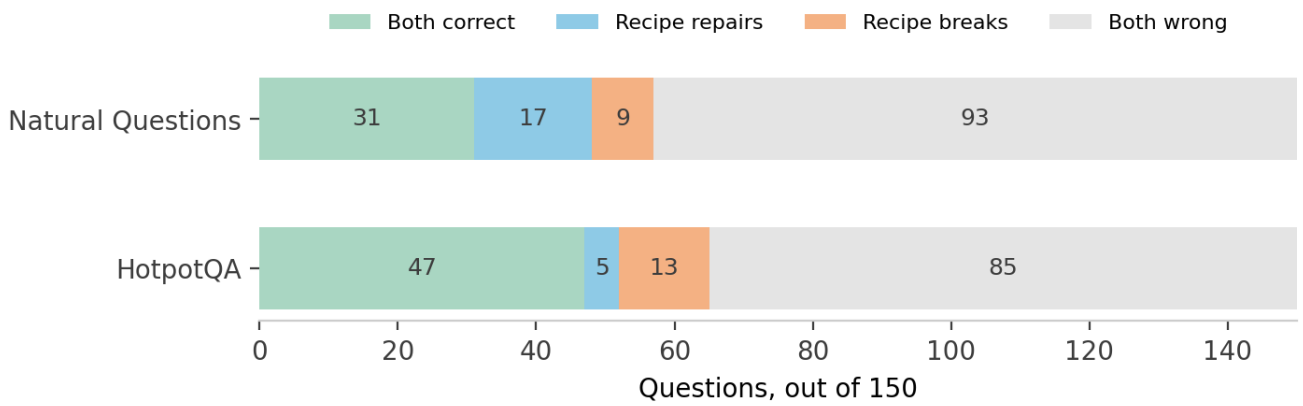


Figure 3: One-shot retrieval against the learned recipe. Each bar is 150 questions. HotpotQA is 47 both correct, 5 repaired, 13 broken, and 85 both wrong. Natural Questions is 31, 17, 9, and 93.

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